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Spectral density-based statistical measures for image sharpness

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Abstract
In industrial applications, such as automated on-line semiconductor production, there is a growing realization of the need for the development of a procedure for periodic performance testing of the image sharpness of the scanning electron microscope. Two statistical and comparative measures of image sharpness based on the generalized variance and bivariate kurtosis of the spectral density for a two-dimensional stationary process are proposed.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Fully automated or semi-automatic scanning electron microscopes (SEM) are now commonly used in semiconductor production and other forms of manufacturing. It is required that these automated instruments be routinely capable of 3 nm or better resolution below 1 kV accelerating voltage, for the measurement of nominal 70 nm to 150 nm size parts of integrated circuits. The testing and proof that the instrument is performing at this level of production on a day-by-day basis, however, has not been routinely done. Once a human operator is no longer monitoring the instrument's performance and where multiple instruments are concerned, an objective diagnostic procedure must be implemented to ensure data and measurement fidelity.

Digital images are composed of individual image points called pixels. Any image taken in a SEM is the result of the interaction of the sample and the electron probe. The sharpness concept described by Postek and Vladar [1] is a practical approach to the evaluation of instrument resolution. A degradation of the sharpness of the image of a suitable test object can serve as one of, perhaps, several indicators of the need for maintenance. Postek and Vladar [2] proposed a procedure based on the image sharpness and it has subsequently been refined into a user-friendly, stand-alone analysis system (Vladar *et al* [3]). The procedure was based on the objective characterization of the spatial Fourier transform of the SEM images of a test object. It was observed that when an SEM image is visually sharper than a second image, the high spatial frequency components of the first image are larger than those of the second. The following is an illustrative example. Figure 1(*a*) shows the performance of a cold field emission SEM on a heavily gold-coated oxide test sample at low accelerating voltage. This micrograph was taken following a poorly implemented maintenance procedure. The image appears to be less sharp and lacking in resolution when compared with a similar micrograph (figure 1(*c*)) taken when the same instrument was operating optimally. Figure 1(*a*) appears to be far less sharp than figure 1(*c*). Figures 1(*b*) and (*d*) show the magnitude distributions of the two-dimensional Fourier transforms or periodograms at spatial Fourier frequencies of the images in figures 1(*a*) and (*c*), respectively. From these figures it is clear that for the sharper image in figure 1(*c*), its cone in figure 1(*d*), which represents the periodogram of the image, is wider than that of the image in figure 1(*b*).

A statistical measure based on the bivariate kurtosis was proposed by Zhang *et al* [4,5] to measure the sharpness of the SEM images. This paper will discuss the statistical issues of a more general approach in greater detail. With this approach, an SEM image can be treated as a realization of a two-dimensional stochastic process. In section 2, we will discuss the autocorrelation structures and the corresponding spectra for the two-dimensional stationary processes. In section 3, we will discuss the moments of multivariate distributions including generalized variance and multivariate kurtosis and their related properties. In section 4, generalized variances and bivariate kurtosis are applied to the two-dimensional periodograms

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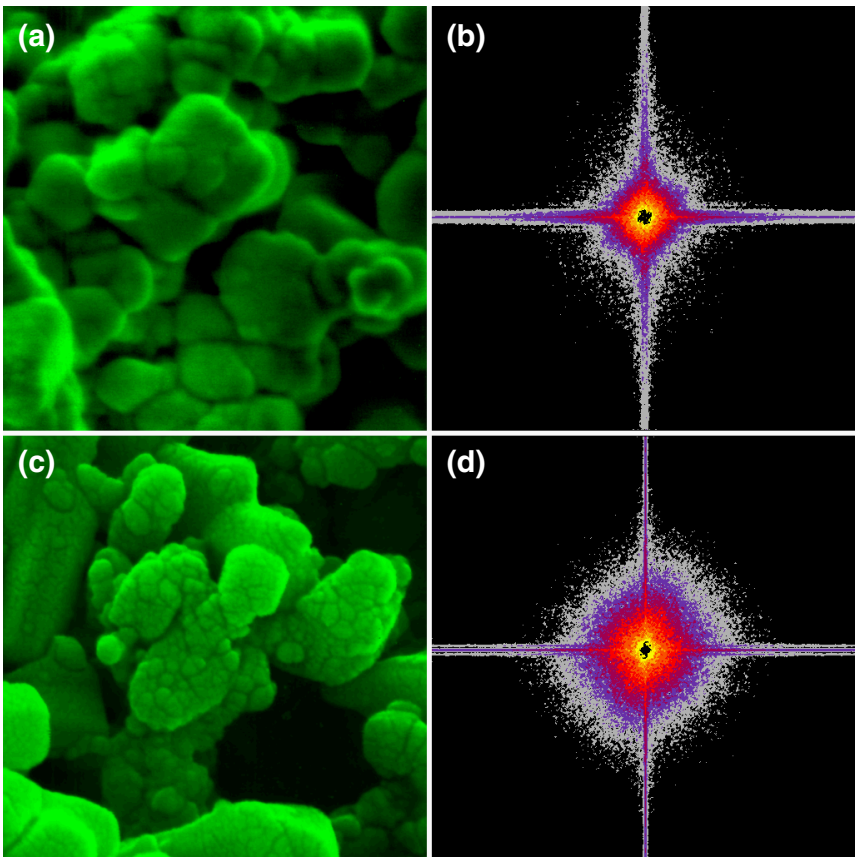

Figure 1. Two SEM images and their two-dimensional periodograms.

of SEM images to measure the image sharpness and a real example is presented, followed by the conclusions in section 5.

2. Two-dimensional stationary processes

We consider a continuous two-dimensional stochastic process $\{X_{t,\tau}\}$, where $-\infty < t, \tau < \infty$. The process $\{X_{t,\tau}\}$ is called weakly stationary if

(1) $\mathrm{E}[X_{t,\tau}] = \mu_X$, a constant for all t and τ.
(2) $\mathrm{Var}[X_{t,\tau}] = \sigma_X^2$, a constant for all t and τ. (1)
(3) The (two-dimensional) autocovariance function $\mathrm{Cov}[X_{t,\tau}, X_{t+s,\tau+u}]$ only depends on lags s and u and is thus denoted by $R(s, u)$.

The indices t and τ can be for time and one-dimensional location or for two spatial parameters in this paper. We assume that all the processes are real-valued. For example, a two-dimensional image process is a two-dimensional process formed by the image points with t and τ as the two spatial coordinates.

Similar to the one-dimensional case, $R(s, u) = R(-s, -u)$ for all s and u. Obviously, $\sigma_X^2 = R(0, 0)$. The two-dimensional autocorrelation is defined as

$$\rho(s, u) = \frac{R(s, u)}{\sigma_X^2}.$$

Similar to the one-dimensional case (e.g. see Priestley [6], p 720), the two-dimensional power spectral density function $f(\omega, \nu)$ is given by

$$f(\omega, \nu) = \left(\frac{1}{2\pi}\right)^2 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \rho(s, u)\mathrm{e}^{-\mathrm{i}s\omega - \mathrm{i}u\nu}\,\mathrm{d}s\,\mathrm{d}u \qquad (2)$$

for $-\infty < \omega, \nu < \infty$. The function $f(\omega, \nu)$ exists for all ω and ν if

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |\rho(s, u)|\,\mathrm{d}s\,\mathrm{d}u < \infty.$$

Obviously, $f(-\omega, -\nu) = f(\omega, \nu)$. Similar to the one-dimensional case (see Priestley [6], p 217), the spectral density function $f(\omega, \nu)$ has the following properties:

(1) $\displaystyle\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\omega, \nu)\,\mathrm{d}\omega\,\mathrm{d}\nu = 1,$

(2) $f(\omega, \nu) \geqslant 0,$

(3) $f(-\omega, -\nu) = f(\omega, \nu)$ for all ω and ν. (3)

Based on the properties in 1 and 2, the spectral density function $f(\omega, \nu)$ is like a two-dimensional probability density function of a bivariate random vector. The normalized integrated spectrum

$$F(\omega, \nu) = \int_{-\infty}^{\omega}\int_{-\infty}^{\nu} f(\xi, \psi)\,\mathrm{d}\xi\,\mathrm{d}\psi$$

is like a bivariate cumulative probability distribution function.

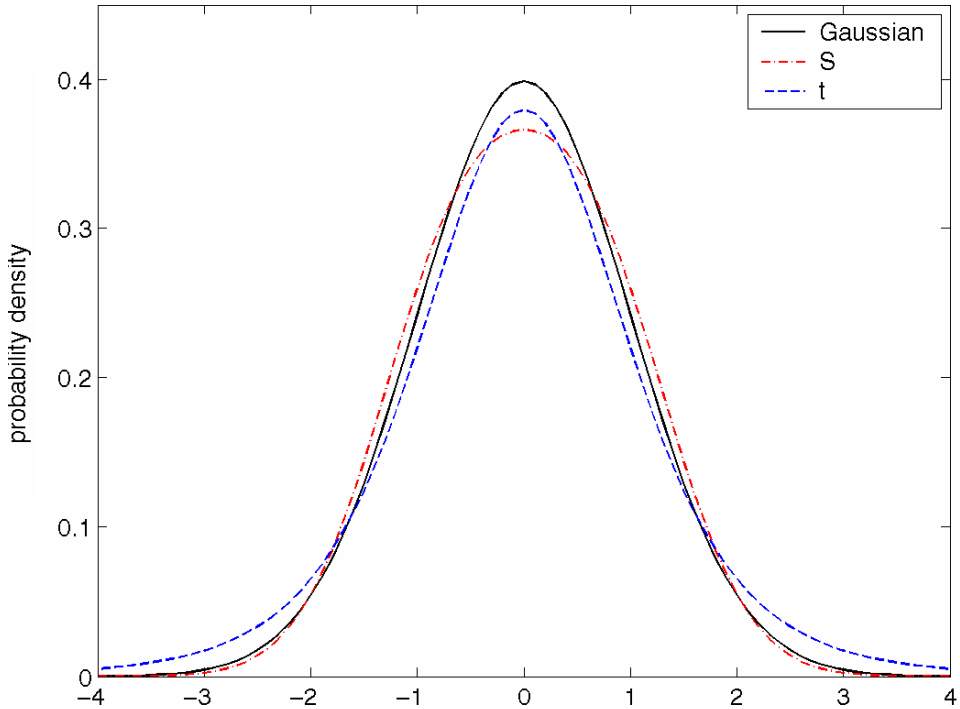

Figure 2. Probability density functions of three distributions.

3. Moments of a multivariate distribution

In this section, we will discuss the measures based on the second and fourth moments of a random vector.

A measure of spread or dispersion in a univariate distribution is the variance. In the multivariate case, an obvious analogue is the covariance matrix. Wilks [7] introduced the determinant of the covariance matrix as a scalar measure of dispersion, which is called the generalized variance. Let W be a p-dimensional random vector with finite moments up to the fourth. Let μ_W be the mean vector and Σ_W be the covariance matrix of W. The generalized variance is $|\Sigma_W|$. In particular, when $p = 2$, $W = (W_1, W_2)'$ and

$$\Sigma_w = \begin{bmatrix} \sigma^2_{W_1} & \rho_{12}\sigma_{W_1}\sigma_{W_2} \\ \rho_{12}\sigma_{W_1}\sigma_{W_2} & \sigma^2_{W_2} \end{bmatrix},$$

where ρ_{12} is the correlation between W_1 and W_2. The generalized variance for W is

$$GV_W = \begin{vmatrix} \sigma^2_{W_1} & \rho_{12}\sigma_{W_1}\sigma_{W_2} \\ \rho_{12}\sigma_{W_1}\sigma_{W_2} & \sigma^2_{W_2} \end{vmatrix} = \sigma^2_{W_1}\sigma^2_{W_2}(1-\rho^2_{12}). \tag{4}$$

We now discuss the kurtosis for univariate and multivariate distributions. For a univariate random variable Z with mean μ_Z and finite moments up to the fourth, the kurtosis is defined as

$$\beta_2 = \frac{\mathrm{E}[(Z-\mu_Z)^4]}{\{\mathrm{E}[(Z-\mu_Z)^2]\}^2}. \tag{5}$$

Assuming the probability density function $f_Z(x)$ $(-\infty < x < \infty)$ exists, the kth central moment is

$$m_k = \mathrm{E}[(Z-\mu_Z)^k] = \int_{-\infty}^{\infty} (x-\mu_Z)^k f_Z(x)\,\mathrm{d}x.$$

Obviously,

$$\beta_2 = \frac{m_4}{(m_2)^2}. \tag{6}$$

It is well known that for any univariate random variable with a Gaussian distribution, $\beta_2 = 3$. The value of β_2 for a random variable can be compared with 3 to determine whether its distribution is 'peaked' or 'flat-topped' relative to a Gaussian distribution as stated in Finucan [8]. Four separate density functions with zero mean and unit variance were compared by Kaplansky [9] to illustrate this property of kurtosis. His results show that among these four distributions with the same variance, the smaller the kurtosis, the flatter the top of the distribution. Figure 2 shows the probability density functions of the standard Gaussian distribution (solid line), the Student-t distribution with degrees of freedom of 5 (dashed line) and the density function $S(x)$ (dotted line) discussed in Kaplansky [9]. The t distribution of degrees of freedom of 5 has kurtosis of 9 (see Johnson *et al* [10], p 366). The distribution with the density function of $S(x)$ proposed by Kaplansky [9] has a kurtosis of 2.667. It is clear that the distribution with a smaller kurtosis is more flat-topped or has a larger shoulder than that with a larger kurtosis. The distribution with the density function of $S(x)$, proposed by Kaplansky [9], with $\beta_2 = 2.667$ has the largest shoulder among the three distributions.

Multivariate kurtosis was proposed by Mardia [11]. Let W be a p-dimensional random vector with finite moments up to the fourth. Let μ_W be the mean vector and Σ_W be the covariance matrix of W. The kurtosis of W is defined as

$$\beta_{2,p} = \mathrm{E}\{(W-\mu_W)'\Sigma^{-1}(W-\mu_W)\}^2. \tag{7}$$

When $p = 1$, $\beta_{2,1}$ becomes the univariate kurtosis or β_2, in (11). When $p = 2$, from Mardia [11] the bivariate kurtosis

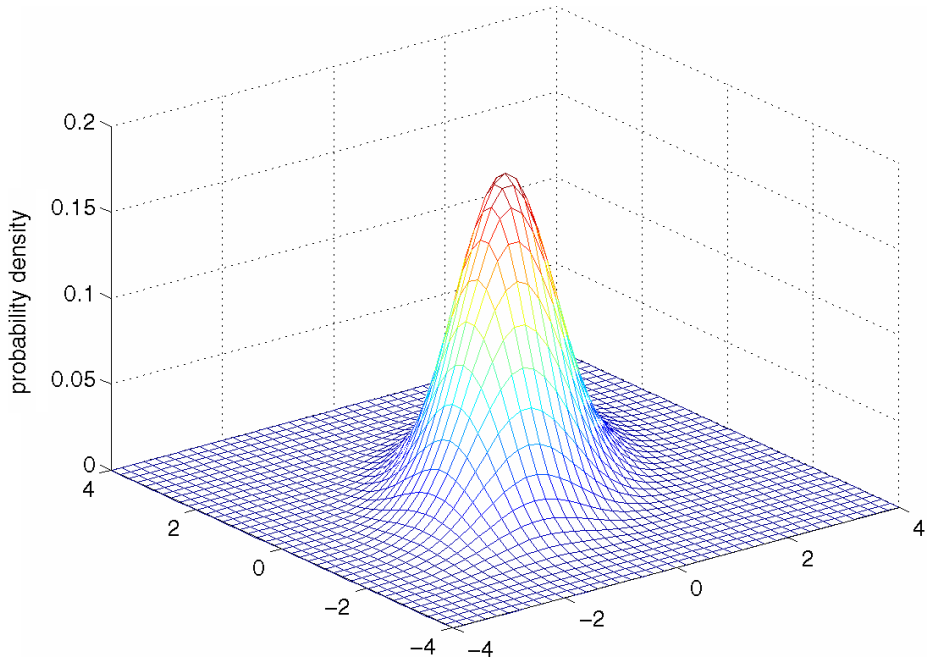

Figure 3. Probability density function of a two-dimensional Gaussian distribution.

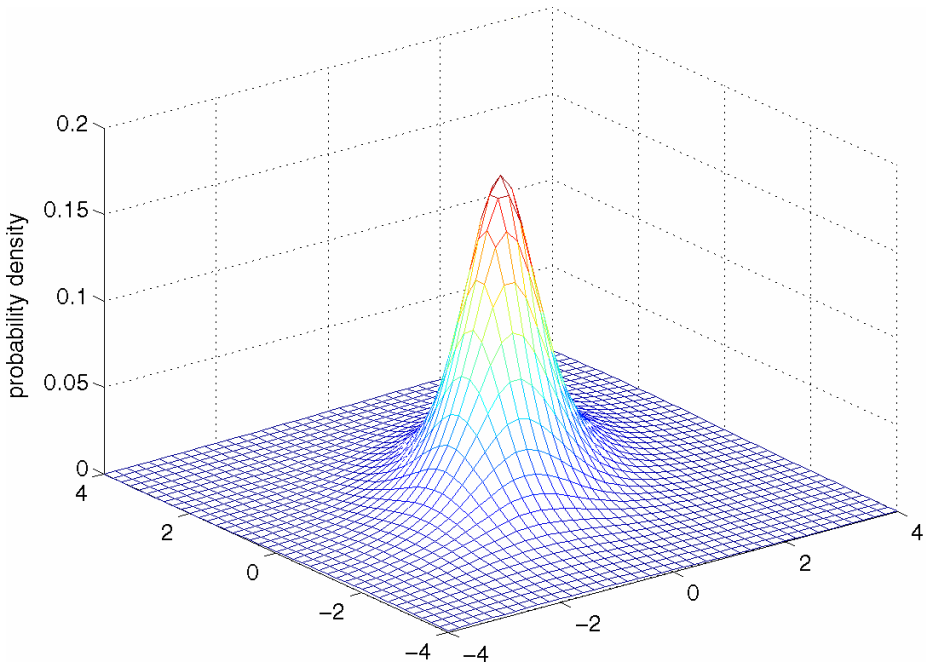

Figure 4. Probability density function of a two-dimensional t distribution.

for $W = (W_1, W_2)'$ is calculated by

$$\beta_{2,2} = \frac{\gamma_{4,0} + \gamma_{0,4} + 2\gamma_{2,2} + 4\rho_{12}(\rho_{12}\gamma_{2,2} - \gamma_{1,3} - \gamma_{3,1})}{(1 - \rho_{12}^2)^2}, \quad (8)$$

where ρ_{12} is the correlation between W_1 and W_2, and $\gamma_{k,l}$ for $k, l = 0, 1, 2, 3, 4$, is calculated by

$$\gamma_{k,l} = \frac{\mathrm{E}[(W_1 - \mu_{W_1})^k (W_2 - \mu_{W_2})^l]}{\sigma_{W_1}^k \sigma_{W_2}^l}. \quad (9)$$

μ_{W_1}, μ_{W_2}, σ_{W_1} and σ_{W_2} are the marginal means and the marginal standard deviations of W_1 and W_2, respectively. In particular, when W is a bivariate Gaussian random vector, the corresponding bivariate kurtosis $\beta_{2,2} = 8$. Figure 3 shows the probability density function of a standardized bivariate Gaussian random vector with $\rho_{12} = 0.5$. In figure 4, a probability density function of a bivariate central t distribution with a degree of freedom of 9 was generated based on equation (1) in Johnson and Kotz [12], p 134. The bivariate t distribution is a joint distribution of $W_i = X_i\sqrt{\nu}/S$ $(i = 1, 2)$, where X_1 and X_2 have a joint standardized bivariate Gaussian distribution with correlation of X_1 and X_2 equal to $\rho_{X_1X_2}$. The $\rho_{X_1X_2}$ in the bivariate t distribution in figure 4

equals 0.5. The random variable S in the definition of W_i is independent of X_i $(i = 1, 2)$ and is distributed as χ_ν with a degree of freedom of ν. The product moments of W_1 and W_2 are calculated and expressed by Gamma functions as follows:

$$\begin{aligned} \mathrm{E}[W_1^k W_2^l] &= \nu^{(k+l)/2}\mathrm{E}[X_1^k X_2^l]\mathrm{E}[\chi_\nu^{-(k+l)}] \\ &= \left(\frac{\nu}{2}\right)^{(K+l)/2} \mathrm{E}[X_1^k X_2^l]\frac{\Gamma((\nu-k-l)/2)}{\Gamma(\nu/2)} \end{aligned} \tag{10}$$

provided $k + l < \nu$. The second equality holds because

$$\mathrm{E}[\chi_\nu^{-(k+l)}] = \left(\frac{1}{2}\right)^{(k+l)/2} \frac{\Gamma((\nu-k-l)/2)}{\Gamma(\nu/2)}.$$

The moment $\mathrm{E}[X_1^k X_2^l]$ $(k, l = 0, 1, 2, 3, 4$ and $k + l < 4)$ can be obtained from Patel and Read [13], p 309. The correlation between W_1 and W_2 is also $\rho_{X_1X_2}$. The kurtosis of the bivariate t distribution, $\beta_{2,2}$, can be obtained from (8). It is easy to show that for a bivariate t distribution, when $\nu > 4$, $\beta_{2,2}$ is independent of $\rho_{X_1X_2}$ and depends only on ν. In fact,

$$\beta_{2,2} = 8\frac{\Gamma((\nu-4)/2)\Gamma(\nu/2)}{\{\Gamma((\nu-2)/2)\}^2}.$$

In this case, the kurtosis of the bivariate t distribution with a degree of freedom of 9 is $\beta_{2,2} = 11.2$. Figures 3 and 4 show that the curvature of the two-dimensional Gaussian distribution with a smaller bivariate kurtosis (=8) has a larger shoulder than that for the bivariate t distribution with a degree of freedom of 9, which has a larger bivariate kurtosis (=11.2).

For the marginal distributions of a bivariate random vector $W = (W_1, W_2)'$, the corresponding marginal variances are $\sigma^2_{W_1}$ and $\sigma^2_{W_2}$. The marginal kurtoses, denoted by β_{2,W_1} and β_{2,W_2}, are the univariate kurtoses for W_1 and W_2, respectively. In fact, from (9) and (5),

$$\beta_{2,W_1} = \gamma_{4,0} \qquad \text{and} \qquad \beta_{2,W_2} = \gamma_{0,4}. \tag{11}$$

The marginal kurtoses can be used to measure the shapes of the shoulders of the corresponding marginal distributions. Marginal variances and marginal kurtoses are used in detecting the astigmatism and instrument vibrations. We will discuss this application in the next section.

4. Use of generalized variance and bivariate kurtosis to measure image sharpness

As mentioned in section 2, a two-dimensional spectral density is like a probability density function. In section 3, we proposed generalized variances to measure the dispersion of a two-dimensional probability density function. In section 3, we also showed that the bivariate kurtosis is a measure of the shoulder of a two-dimensional probability density function. Thus, for a given two-dimensional spectral density, the corresponding generalized variances and bivariate kurtosis can be used to measure the dispersion and the shape of the shoulder of the two-dimensional spectral density.

For a two-dimensional stationary process $\{W_{t,\tau}\}$, its spectral density $f(\omega, \nu)$ is usually estimated by its two-dimensional discrete realizations. For a given set of discrete two-dimensional lattice data, $\{w_{t,\tau}; t, \tau = 1, 2, \ldots, N\}$ the two-dimensional discrete Fourier transform is

$$\xi(\omega, \nu) = \frac{1}{2\pi N}\sum_{t=1}^{N}\sum_{\tau=1}^{N} w_{t,\tau}\,\mathrm{e}^{-\mathrm{i}t\omega-\mathrm{i}\tau\nu} \tag{12}$$

for $\omega, \nu = 2\pi\lambda/N$; $-[(N-1)/2] \leqslant \lambda \leqslant [N/2]$, where $[x]$ denotes the largest integer not exceeding x. The corresponding two-dimensional periodogram (see Koopmans [14], p 258 and Priestley [6], p 722)

$$I(\omega, \nu) = |\xi(\omega, \nu)|^2 = \frac{1}{(2\pi N)^2}\left|\sum_{t=1}^{N}\sum_{t=1}^{N} w_{t,\tau}\,\mathrm{e}^{-\mathrm{i}t\omega-\mathrm{i}\tau\nu}\right|^2 \tag{13}$$

is an estimator of the two-dimensional spectral density. It is well known that for any given ω and ν, the expectation of $I(\omega, \nu)$ approaches $f(\omega, \nu)$ when N goes to infinity (Priestley [6], pp 722–3). That is, $I(\omega, \nu)$ is an asymptotically unbiased estimator of the two-dimensional spectral density. The frequencies of $\omega, \nu = 2\pi\lambda/N$; $-[(N-1)/2] \leqslant \lambda \leqslant [N/2]$ are called Fourier frequencies (e.g. see Ripley [15], p 80). The corresponding $\{I(\omega, \nu)\}$ are the values of the periodogram at these Fourier frequencies. For simplicity we denote these frequencies by ω_i, ν_i $(i = 1, 2, \ldots, N)$.

For a discrete probability distribution, the bivariate kurtosis is calculated by the formulae in section 3. A discrete density based on $\{I(\omega_i, \nu_j)\}$, $i, j = 1, 2, \ldots, N$ is obtained by normalization. Namely,

$$h(\omega_i, \nu_j) = \frac{I(\omega_i, \nu_j)}{\sum_{m=1}^{N}\sum_{n=1}^{N} I(\omega_m, \nu_n)}, \tag{14}$$

for $i, j = 1, 2, \ldots, N$. The corresponding marginal means and variances of $h(\omega, \nu)$ are calculated by

$$\begin{aligned} \mu_\omega &= \sum_{i=1}^{N}\omega_i\sum_{j=1}^{N}h(\omega_i, \nu_j), \\ \mu_\nu &= \sum_{j=1}^{N}\nu_j\sum_{i=1}^{N}h(\omega_i, \nu_j) \end{aligned} \tag{15}$$

and

$$\begin{aligned} \sigma_\omega^2 &= \sum_{i=1}^{N}(\omega_i-\mu_\omega)^2\sum_{j=1}^{N}h(\omega_i, \nu_j), \\ \sigma_\nu^2 &= \sum_{j=1}^{N}(\nu_j-\mu_\nu)^2\sum_{i=1}^{N}h(\omega_i, \nu_j). \end{aligned} \tag{16}$$

The covariance between ω and ν is

$$\sigma_{\omega\nu} = \sum_{i=1}^{N}\sum_{j=1}^{N}(\omega_i-\mu_\omega)(\nu_j-\mu_\nu)h(\omega_i, \nu_j). \tag{17}$$

The correlation between ω and ν is

$$\rho_{\omega\nu} = \frac{\sigma_{\omega\nu}}{\sigma_\omega\sigma_\nu}. \tag{18}$$

Then from (17) the corresponding $\gamma_{k,l}$ for $k, l = 0, 1, 2, 3, 4$ is

$$\gamma_{k,l} = \frac{\sum_{j=1}^{N}\sum_{i=1}^{N}h(\omega_i, \nu_j)(\omega_i-\mu_\omega)^k(\nu_j-\mu_\nu)^l}{\sigma_\omega^k\sigma_\nu^l}. \tag{19}$$

The generalized variance corresponding to $h(\omega, \nu)$ can be obtained by (4) and (14)–(18). The bivariate kurtosis corresponding to $h(\omega, \nu)$ can be obtained by (8) and (19). Therefore, for a given image, the corresponding generalized variances and bivariate kurtosis are calculated based on the two-dimensional periodogram of the image.

As we observed in the example presented in the introduction, of the sharp image in figure 1(*c*) and its cone in figure 1(*d*), the cone in figure 1(*d*) is wider than the cone of the image in figure 1(*a*). Thus, the generalized variance proposed in (4), which is a measure of the dispersion of the magnitude distributions or cones in the example, can be used to measure image sharpness. Obviously, a sharper image such as the one in figure 1(*c*) that corresponds to a larger generalized variance has a larger dispersion of the magnitude distribution as in figure 1(*d*).

When the beam of the SEM scans the sample, the zero spatial frequency component ($\omega, \nu = 0$) corresponds to the average grey level of the image. As we move away from the origin of the frequencies, the low frequencies correspond to the slowly varying components of the image or the large features. When we move further away from the origin, the higher frequencies begin to correspond to faster grey level changes in the image, which give data of finer details such as the edges. On the other hand, very high frequencies corresponding to abrupt changes in grey level comprise the noise. (See the discussions in Gonzalez and Woods [16], pp 156–7.) An image of a sample, which has fine details at a given magnification, is sharper if there are more high frequency (but not too high) changes in it. Russ [17], pp 306–7, used an example to show that 'The frequency transform of an image can be used to optimize focus and astigmatism. When an image is in focus, the high-frequency information is maximized in order to sharply define the edges. This provides a convenient test for sharpest focus.' In that sense, kurtosis, which measures the width of the shoulder of the distribution corresponding to the strength at high frequencies, but not too high, is used to measure image sharpness. Hence, when the bivariate kurtosis of the (normalized) periodogram corresponding to one image is smaller than the periodogram of the second image, the first image is sharper than the second one. The example in the introduction shows the relationship between the image sharpness and the width and shape of the bivariate periodogram. An SEM image is represented by a set of discrete two-dimensional lattice data. The corresponding two-dimensional periodogram with the zero frequency at the centre is calculated from (13). From the discussion in section 3, between two SEM images, a sharper image that corresponds to a smaller bivariate kurtosis of the periodogram has a larger shoulder or a flatter shape.

The marginal variances of the marginal distribution of ω and ν are obtained from (16). From (11), the marginal kurtoses of the marginal distributions of ω and ν are $\beta_{2,\omega} = \gamma_{4,0}$ and $\beta_{2,\nu} = \gamma_{0,4}$. The marginal variances are used to measure the dispersions of the marginal distributions. The difference between the two marginal variances measures the difference between the dispersions of the marginal distributions. The marginal kurtoses are used to measure the shapes of the shoulders of the two marginal distributions. The difference between the marginal kurtoses measures the difference between the widths of the shoulders of the marginal distributions. When astigmatism is present, the two-dimensional power spectrum is asymmetric (see, e.g., Russ [17], pp 306–7). Therefore, the marginal variances and marginal kurtoses can be used to detect the asymmetry of the periodogram in vertical and horizontal sections corresponding to the astigmatism or misalignment in the image, and possible instrument vibration. We use

$$r_{\mathrm{V}} = \frac{|\sigma_\omega^2 - \sigma_\nu^2|}{\min(\sigma_\omega^2, \sigma_\nu^2)} \tag{20}$$

and

$$r_{\mathrm{K}} = \frac{|\beta_{2,\omega} - \beta_{2,\nu}|}{\min(\beta_{2,\omega}, \beta_{2,\nu})} \tag{21}$$

to measure the relative differences between the marginal variances and the marginal kurtoses, respectively.

The following example demonstrates how the generalized variance and bivariate kurtosis are used to measure the sharpness of SEM images. The sample is an etched silicon wafer with the artefact referred to as 'grass', which is an etching artifact that can occur on silicon wafers during processing, as stated in Postek [18]. The considerations on the sample structure are as follows: if the sample is perfectly flat, with no details, the measurements will not report any meaningful information. An ideal sample would have no 'signature' in the Fourier domain, which in this case would not have any central peak. As practical staples, gold-on-carbon targets might work very well. The grass sample was chosen for its resiliency, compatibility with silicon integrated circuit technologies, easy cleaning, possibility for contamination rate measurements, etc. It does have a non-negligible impact on the results, but as a comparative measurement target it works fine. In figure 5, a series of five micrographs shows a representative set of experiments developed to demonstrate the sharpness analysis procedure. All the micrographs were taken by the same instrument with only one parameter changed for each image. Figure 5(*a*) is a micrograph of the grass sample taken with the SEM conditions set for best focus images. Figure 5(*b*) is a similar micrograph also under best focus condition. Figure 5(*c*) is a micrograph, which is not properly focused. Figure 5(*d*) is a micrograph with intentional astigmatism in the horizontal direction. Figure 5(*e*) is a micrograph also with intentional astigmatism, but in the vertical direction.

The second row in figure 5 shows the magnitude distributions of the two-dimensional periodograms corresponding to the images in figures 5(*a*)–(*e*). Darker points mean high magnitudes of the values of a periodogram. Each two-dimensional periodogram is calculated based on a sampling of 1024 by 1024 Fourier frequency points. The third row in figure 5 shows the same magnitude distributions as in the second row, after some image processing (and colouring), to show the inner structure of the two-dimensional periodograms better. The fourth and the last rows show the one-dimensional periodograms corresponding to the horizontal and vertical directions of the images in figures 5(*a*)–(*e*), respectively.

Generalized variances and the bivariate kurtoses of the two-dimensional periodograms were calculated by (4) and (8) for the micrographs, respectively, and listed in table 1. The generalized variances are in units of 10^7. Figures 6 and 7 are the graphical representation of the generalized variances

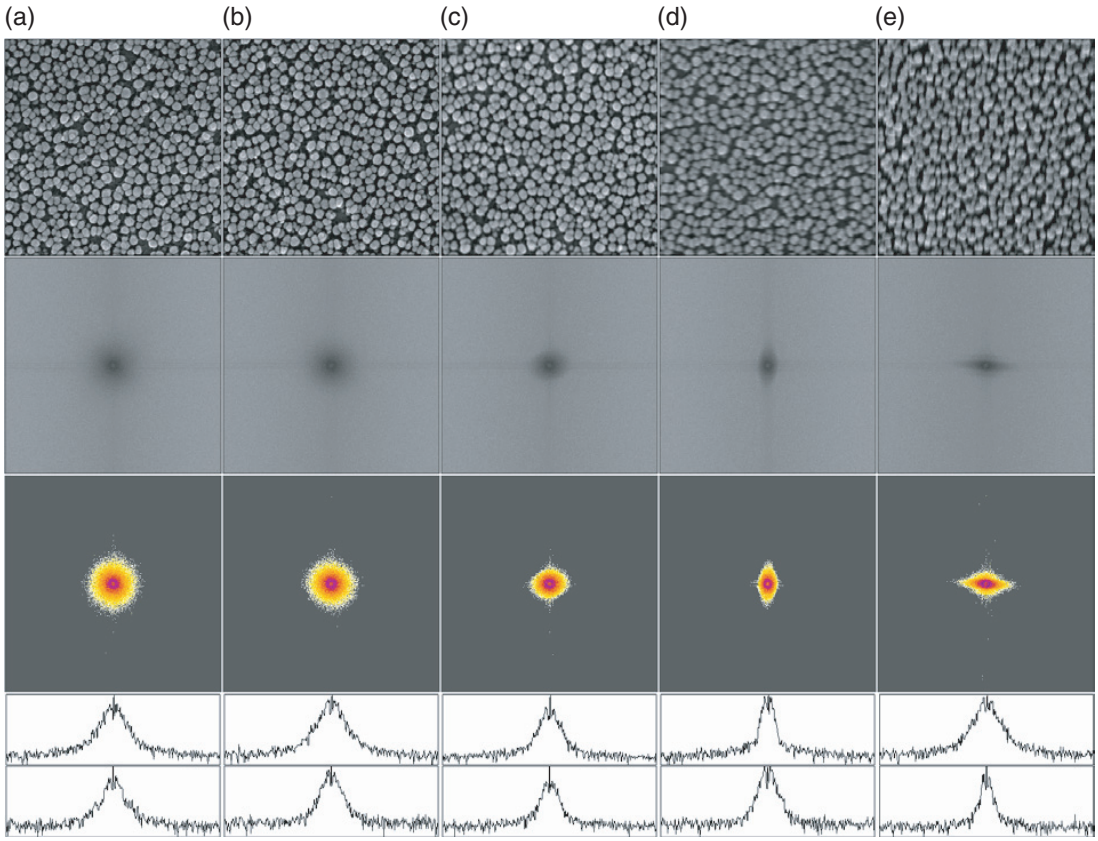

Figure 5. Five micrographs and their two-dimensional and one-dimensional periodograms.

Table 1. Generalized variances, bivariate kurtosis and their relative differences.

Images in figures	$\beta_{2,2}$	$GV(\times 10^7)$	r_K	r_V
5(*a*)	5.9024	1.1266	0.0012	0.0035
5(*b*)	5.9285	1.1075	0.0039	0.0059
5(*c*)	6.0960	1.0353	0.0187	0.0170
5(*d*)	6.1124	1.0582	0.0597	0.0619
5(*e*)	6.0350	1.0887	0.0634	0.0640

(in units of 10^7) and bivariate kurtosis, respectively. As discussed earlier, a high number for generalized variance or a low number for bivariate kurtosis indicates a better quality image or high sharpness. Micrographs in figures 5(*a*) (focused) and 5(*b*) (second focused) are the known good images with generalized variance values of 1.13 (for figure 5(*a*)) and 1.11 (for figure 5(*b*)), and bivariate kurtosis values of 5.90 (for figure 5(*a*)) and 5.93 (for figure 5(*b*)). The image in figure 5(*a*) is slightly sharper than that in figure 5(*b*), which is indicated by a larger generalized variance and a smaller bivariate kurtosis. The visually poorer image in figure 5(*c*) (not focused) exhibits a lower generalized variance value of 1.09 and a higher bivariate kurtosis value of 6.10. The images with astigmatism in figures 5(*d*) and 5(*e*) have lower generalized variances and higher bivariate kurtosis, which indicate that they are not sharp. The generalized variance value for the image in figure 5(*e*) is higher than that in figure 5(*c*), the one not focused, while the bivariate kurtosis value for the micrograph in figure 5(*e*) is lower than that in figure 5(*c*). Does this mean that the image in figure 5(*e*) is sharper than that in figure 5(*c*). We will discuss this issue below. In calculating the bivariate kurtoses, very high frequencies are cut off since they represent noise without image information.

Marginal variances and marginal kurtoses can be used to measure whether an image contains astigmatism. We found that when astigmatism does exist, both the relative difference between the marginal variances and that between the marginal kurtoses corresponding to the horizontal and vertical profiles of the two-dimensional periodogram will show higher values than that which contains no astigmatism. The relative differences of the marginal variances and the marginal kurtosis for the images in figure 5(*a*)–(*e*) are calculated by (20) and (21) and listed in table 1. Figure 8 shows the relative differences of the marginal variances and the marginal kurtoses. The images in figures 5(*d*) and (*e*) demonstrate a much greater relative marginal variance difference ($r_V = 0.06$) and relative marginal kurtosis difference ($r_K = 0.06$) compared with those for the focused images ($r_V = 0.004$ and 0.006, and $r_K = 0.001$ and 0.004) in figures 5(*a*) and (*b*) and the one not focused ($r_V = 0.017$ and $r_K = 0.019$) in figure 5(*c*). The anomalous data set becomes obvious when a study of the relative differences of marginal variances and marginal kurtoses is performed. The high relative differences of marginal variances and/or marginal kurtoses are indicators that something is radically wrong and a service measure should be implemented. Therefore, checking the relative difference of the marginal variances and/or the relative difference of the marginal kurtosis would be the first step for eliminating anomalous data such as those seen in figures 5(*d*) and (*e*).

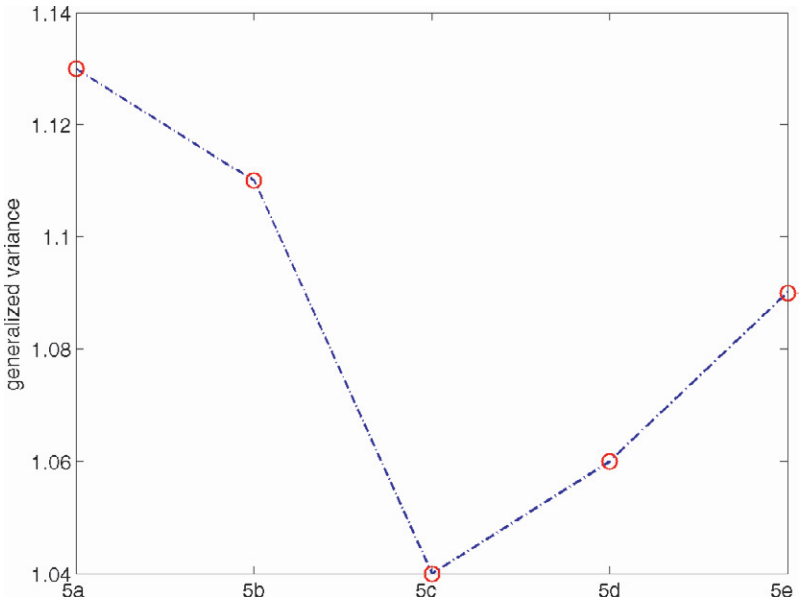

Figure 6. Generalized variances of the five samples.

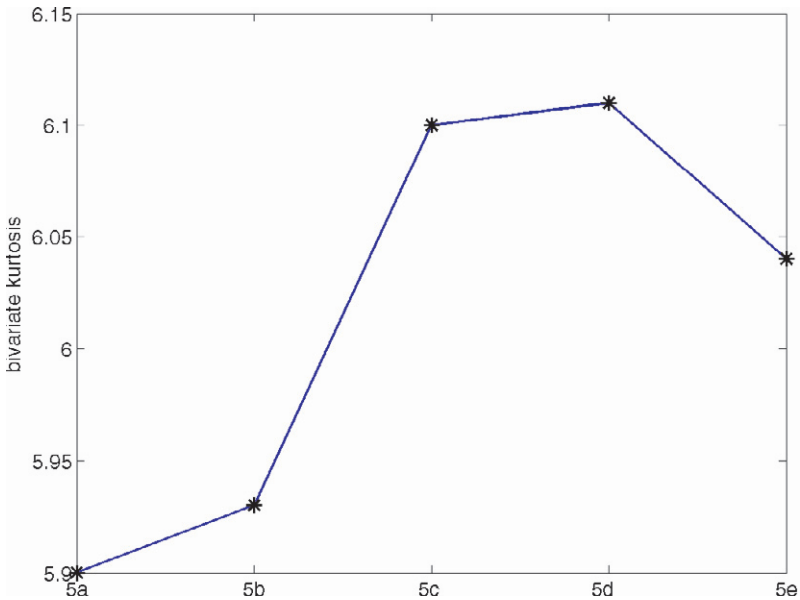

Figure 7. Bivariate kurtosis of the five samples.

We have done limited comparisons between the two image sharpness measures for SEM images based on the generalized variance and the bivariate kurtosis. Both measures work successfully in distinguishing images not focused from those properly focused. In the above example, the generalized variance is a measure with a little higher sensitivity than the bivariate kurtosis. However, using bivariate kurtosis to measure the sharpness of a SEM image has an advantage because the dimensionless kurtosis is insensitive to the scale factor of the image. A detailed discussion of the advantages of using kurtosis in practice can be found in Postek *et al* [19]. Comparing the measures proposed in this paper with the method described in [1–3], we find that both methods work well. The results were found to be repeatable and well tracked. The image sharpness and eccentricity values in [1–3] give clues to the SEM operator for necessary adjustments to be made. Generalized variance and kurtosis are useful and marginal generalized variances and marginal kurtoses give further information for achieving closer-to-ideal electron–optical column settings.

5. Conclusion

While the variance is used to measure the dispersion of a probability distribution, the kurtosis has been used to measure the shape and width of the shoulder of the distribution. These properties are exploited for the spectral density of a two-dimensional stationary process and then successfully applied to SEM as comparative measures, to measure the sharpness of similar images. The generalized variance and

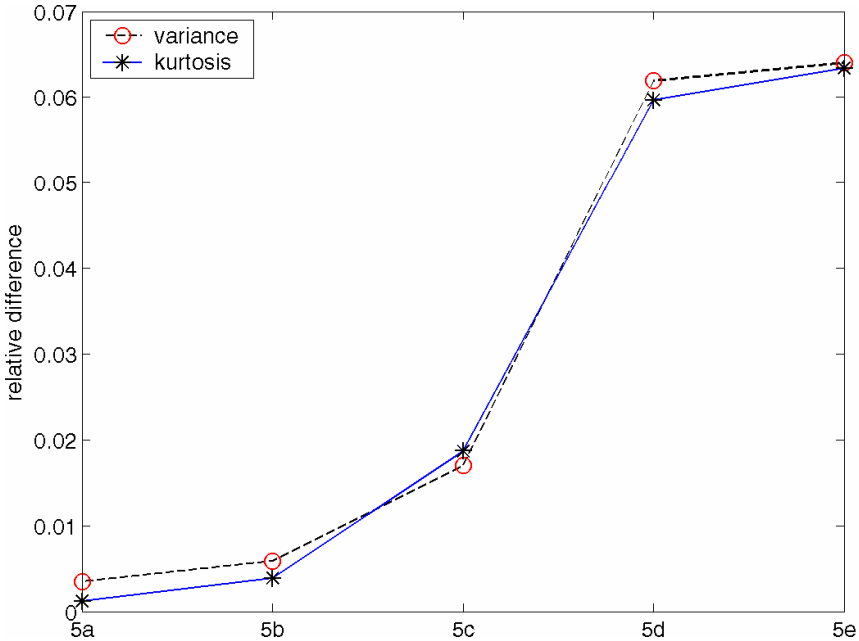

Figure 8. Relative differences between marginal variances and kurtoses.

bivariate kurtosis and the corresponding relative differences of the marginal variances and marginal kurtoses have been implemented in a workstation system and used in the semiconductor industry for SEM performance monitoring. In fact, these approaches can be applied to any image-generating system including SEM to measure image sharpness.

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